CYBERSENTINEL COPILOT

Presentation architecture · user input to final analyst output

Frontend → backend → seven investigation agents → aligned Synapt roles and tools → result → human-reviewed feedback

PHASE 1 · USER INPUT → FRONTEND → BACKEND ENTRY

1 · SOC analyst input
Record ID · question
Attack family · protocol
Optional raw flow

2 · Frontend dashboard
Investigation tab
Search · chat · feedback
HTML/CSS/JavaScript

3 · FastAPI backend
POST /investigate/flow
Route dispatch
Shared services

4 · Pydantic contract
Types · ranges · lengths
At least one input
HTTP 422 on failure

5 · LangGraph entry
Create InvestigationState
Start compiled StateGraph
Carry evidence forward

PHASE 2 · LANGGRAPH INCIDENT INVESTIGATION — SEVEN AGENTS IN EXECUTION ORDER

Agent 1 · Guardrail
guardrails.py
Injection · relevance
Action/telemetry safety

Agent 2 · Flow Triage
sqlite_store.py + RAG
Exact record / 404
Historical similarity

Agent 3 · ML Analysis
predictor.py + anomaly.py
Family · risk · deviations
IsolationForest + MITRE

Agent 4 · Knowledge
synapt_knowledge_agent
Sanitized provider question
Procedures + indicators

Agent 5 · Priority
policy.py
Deterministic P1–P4
Rationale + missing context

Agent 6 · Response
workflow.py builder
Steps · response drafts
Provenance + limitations

Agent 7 · Judge
evaluator.py
Hard safety gate
Grounding/NIST/relevance

PHASE 3 · SYNAPT AGENT ROLES + MCP TOOLS — ATTACHED TO THE CALLING AGENT

Logical roles share one physical client/provider path

Historical Incident RAG Agent
historical-incident-rag-agent
Declared tools:
trident_search · trident_query

Threat Intel & SOP Agent
incident-knowledge-agent
Declared tools:
trident_query · trident_get_procedures

Judge Review Agent
judge-review-agent
Declared tool:
trident_query

Shared SynaptClient → MCP initialize → tools/list → tools/call
Runtime retrieve(): trident_query preferred; trident_search fallback · provider ID injected when tool schema supports it

Synapt Context Substrate · assist_cyber

PHASE 4 · LOCAL DATA, MODELS, POLICY, AND STATE USED BY THE AGENTS

SQLite + FTS5
2,313,810 flows · exact lookup
291 summaries · BM25 keyword retrieval

Local ML models
HistGradientBoosting · 65 features · 9 families
IsolationForest · anomaly threshold · benign baseline

Deterministic intelligence
MITRE tactic map · P1–P4 policy
Investigation/response templates · provenance rules

Feedback + evidence reports
analyst_feedback · candidate model bundle
Anomaly, evaluation, and improvement JSON reports

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PHASE 5 · FINAL OUTPUT FLOW — CONTINUE LEFT TO RIGHT

Connector A: Judge → backend response → frontend → analyst → feedback

10 · Backend mapper
FlowInvestigationResponse
Normalize optional fields
Return JSON

11 · Frontend results
Family · confidence · risk
Anomaly · priority · RAG
MITRE · guidance · verdict

12 · Analyst decision
Verify evidence
Monitor · escalate · dismiss
Manually approve actions

13 · Feedback review
Submit rating/verdict
Human accept/reject gate
Store analyst correction

14 · Improvement evidence
Holdout benchmark
Candidate artifact/report
Reviewed learning input

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SUPPORTING FRONTEND FLOWS AND PROJECT TOOL CHAIN

Incident Search
/search/incidents
SQLite filters + FTS5
+ Historical RAG role

SOC Copilot Chat
/chat
Guardrails + flow/classifier context
+ Knowledge role

Status Panels
/health + /api/synapt/agents
SQLite/model indicators
Three logical role statuses

Offline Preparation
Parquet → SQLite
Train classifier/anomaly
trident_ingest knowledge

Core Tools
FastAPI · Pydantic · LangGraph
SQLite/FTS5 · scikit-learn
Synapt MCP · DeepEval adapter

Presentation accuracy note

The three Synapt names are logical UI roles. Current HistoricalIncidentRAG, knowledge, judge, and chat calls share SynaptClient.retrieve(); trident_get_procedures is displayed as a granted tool but is not directly called by the investigation workflow.

Solid arrows: execution order · dashed arrows: dependency/tool attachment · no automated containment is executed